

Basic Mathematical Operators

Operator	Example	Result
+	12 + 5	17
-	12 - 5	7
*	4 * 7	28
/	16 / 4	4.0
//	16 // 4	4
**	3 ** 4	81

```
print(12 + 5)
print(12 - 5)
print(4 * 7)
print(16 / 4)
print(16 // 4)
print(3 ** 4)
```

Example: `print(16 / 4)`

Output is 4.0, Floating number instead of integer, Python default behavior it's called **Typecasting**.

Example: `print(type(16 / 4))` Output: `<class 'float'>`

Float is a Data Type used to represent and Decimal Points.

Type – Typecasting, **Python is converting the result into Floating point number.**

Double Asterisk: It gives access to Exponents or Powers

Example: `print( 3**4)`

Example: `print(4*6+8/4-9)`

Python Follows a certain order of procedure, meaning some operations Like Divisions or Multiplication Take priority over others, Like Additions or Subtractions.

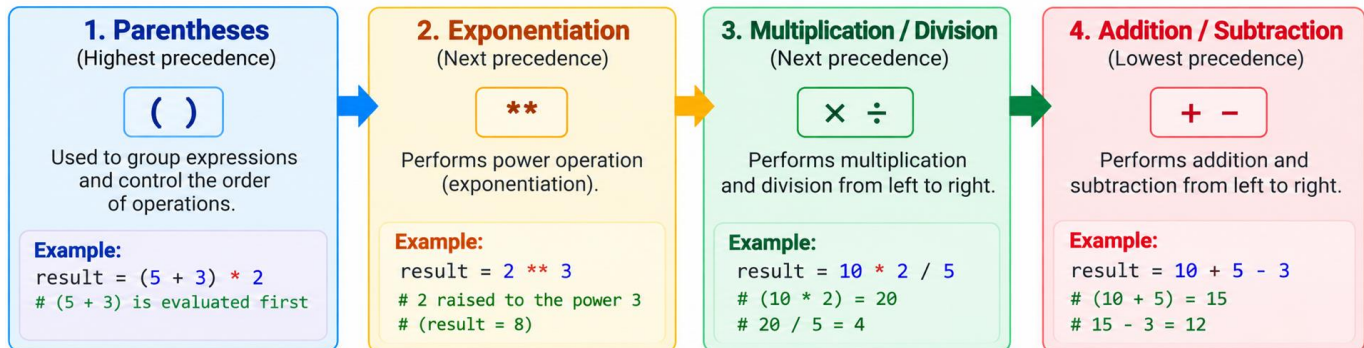
Operator	Meaning	Example	Result
/	Normal division	16 / 5	3.2
//	Floor division (quotient)	16 // 5	3
%	Modulus (remainder)	16 % 5	1

Operator Precedence — PEMDAS

The notes use the familiar **PEMDAS** ordering:

P - Parentheses, E- Exponents, M – Multiplication, D - Division, A – Addition, S- Subtraction.

Operators at the same precedence are generally evaluated from left to right.



```
result = (2 + 3) * 4 ** 2 / 2 - 5
# (2+3) = 5
# 4**2 = 16
# 5*16 = 80
# 80/2 = 40
# 40-5 = 35
```

Order of Execution as Per Priority.

Local and Global Variables

Local & Global Variables are used to store data or information that can be accessed & modified while a program runs.

Example:

```
local_var = 5
print(local_var)
```

```
global_var = 5
print(global_var)
```

Local variable

```
def show_value():
    local_var = 7
    print(local_var)

show_value()
```

A local variable is created within a function and is normally accessible within that function's local scope.

Local variables are temporary & can only be accessed within that specific Function or Block.

Once the Function finishes running, the local Variable is deleted & it's Value is no longer Available or Used.

It exists temporary in a specific area function & does not interface with other parts of the Program.

Global variable

```
global_var = 20

def show_value():
    print(global_var)

show_value()
```

A global variable is defined outside functions and usually at the top of the program.

This type of variable is accessible & modifiable from any part of the program, including inside the functions.

However, to modify a global variable inside a function you need to declare it as Global, Python uses the **“global keyword”**.

```
global_var = 42

def update_value():
    global global_var
    global_var = 100

update_value()
print(global_var) # 100
```

Scope concept

